

Antonio Carmagnani

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Dear Hiring Team at Tetra Tech,

I trained as an architect, and what progressively drew me beyond the drawings were the questions a design studio rarely stops on - why something is being built, whether it should be built there at all, and what its impact will be over its life. Following those questions took me to a Master's Degree in Architecture Computation at The Royal Danish Academy and, within it, specifically toward data, which is the part that makes it possible to answer them with evidence instead of intuition, and from there into sustainability and responsible business consulting. Tetra Tech's Sustainable Building Design team holds those questions where they belong, in the energy a building uses, the carbon embodied in what it is made of, and the decisions taken while there is still time to change them.

During my time in Ramboll Management Consulting's Responsible Business Conduct team I ran human rights due diligence for corporate clients under the UN Guiding Principles and the OECD Guidelines, and in practice a large part of that was data work. I built a fully automated risk assessment system in Excel that consolidated supply chain risk data into a single dashboard which updated salient risks dynamically, an impact assessment tool that scored more than forty risks by severity so the most material ones surfaced first, and a mapping of the supply chains of over twenty critical raw materials under the EU Critical Raw Materials Act. The common task underneath all of it was taking broad and uneven bodies of evidence and structuring them rigorously enough to produce scored, defensible outputs. Sustainability reporting, and the research behind low-carbon materials and renewable options, is that same discipline applied to a building rather than a supply chain.

Material research was at the centre of my master's work at The Royal Danish Academy. I developed a construction technique using blue biomass, seashells and algae bound into a rammed composite, running LCA metrics alongside parametric simulation so that structural performance and environmental impact were pushed in the same iteration rather than traded off afterwards. That work became the material basis for the school's installation at the Venice Architecture Biennale 2025, where I produced the physical panels shown in the seven-metre structure built at the Arsenale with CITA, GXN and 3XN. What the period taught me is that a new material only becomes an engineering proposition once someone has taken it far enough that it has to be built and has to stand up. Embodied carbon studies and circular economy strategy are that work in professional form, and I would rather arrive having already taken a material that far.

Tetra Tech works across the full lifecycle of a project, which is where sustainability stops being an assessment attached at the end and starts shaping what gets built. That is where I want to keep building sustainability work that is measurable, backed by evidence, and able to withstand scrutiny.

Best,

Antonio Carmagnani